
Revealing the activity of different iron carbides for Fischer-Tropsch synthesis

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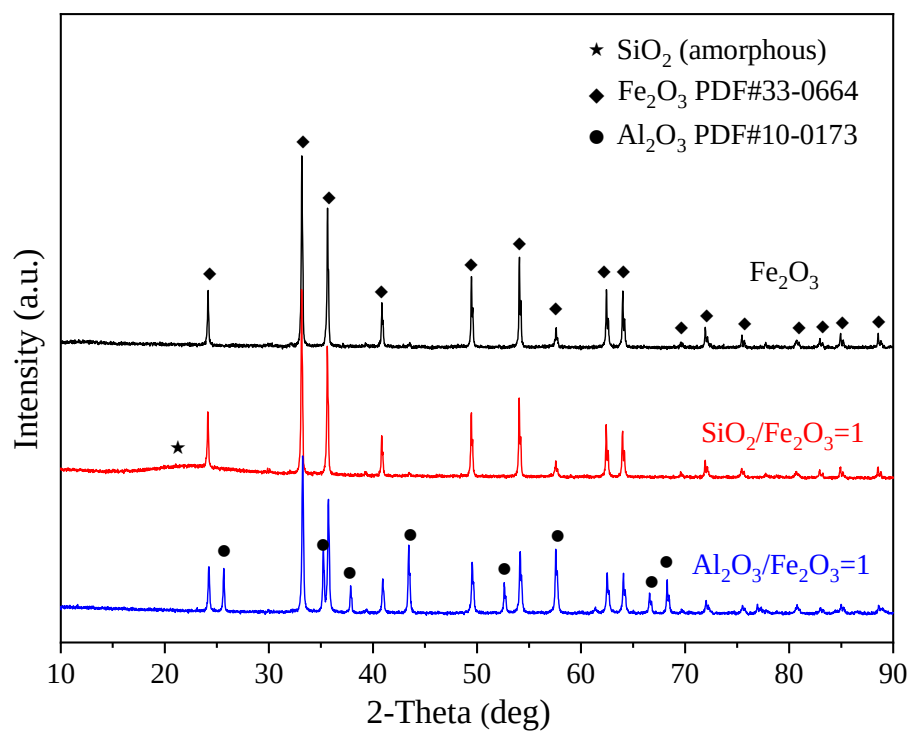


Figure S1. The XRD patterns of fresh catalysts.

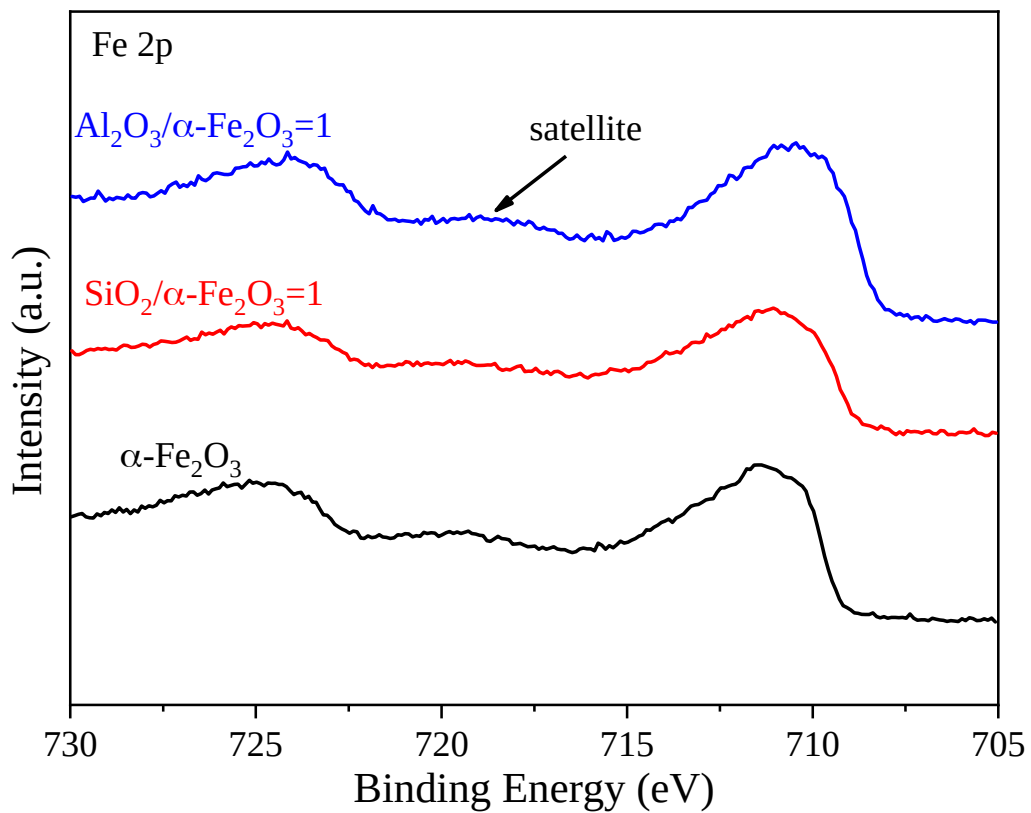


Figure S2. XPS spectra of Fe 2p on the fresh catalysts.

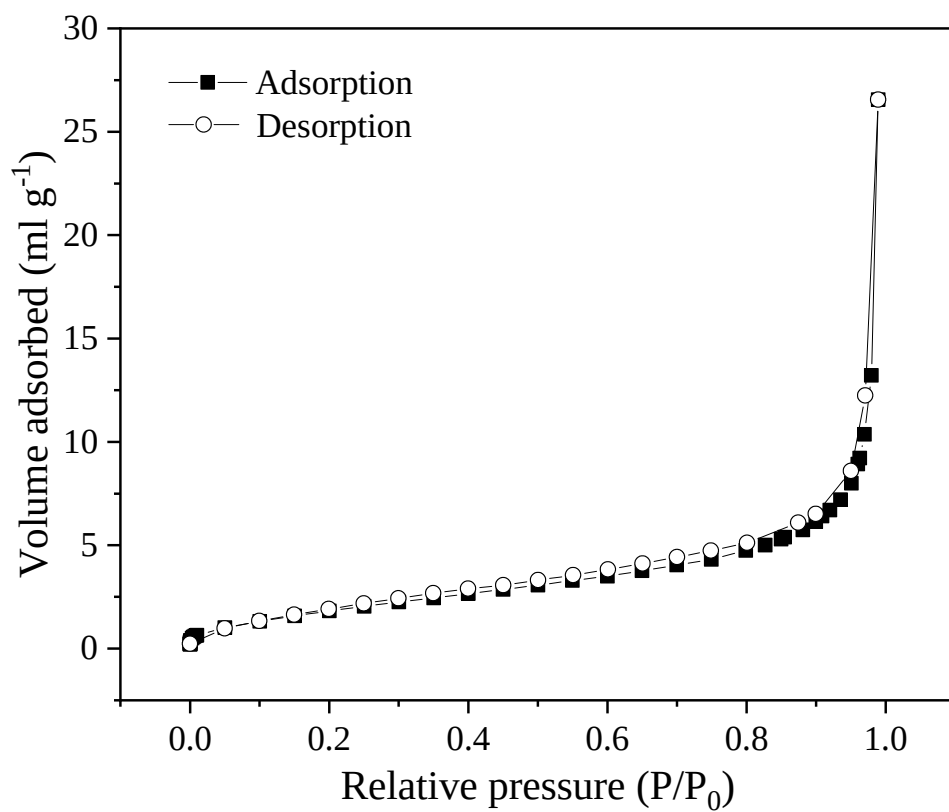


Figure S3. N₂ adsorption-desorption isotherms of the fresh α -Fe₂O₃ catalyst.

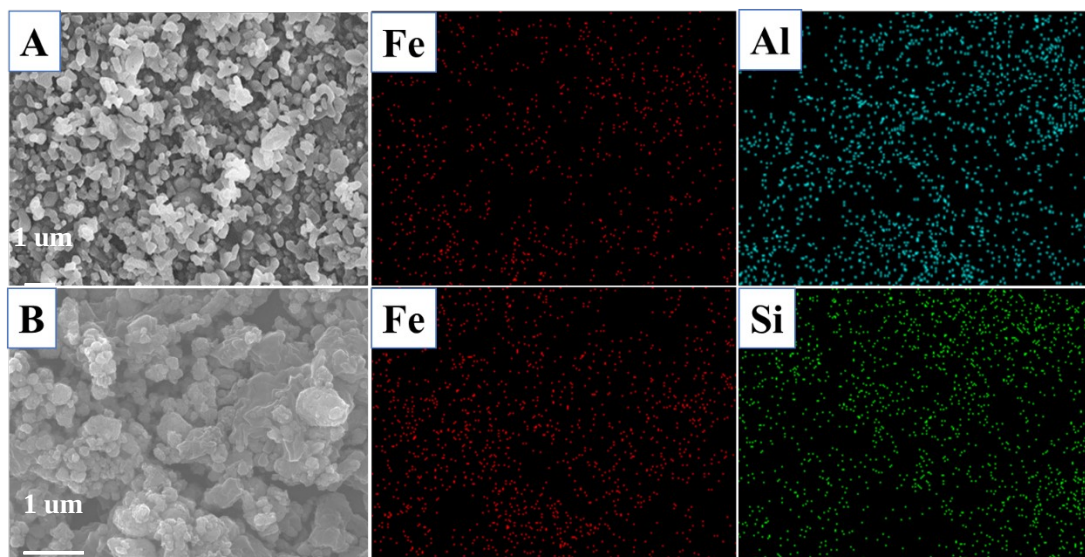


Figure S4 A) SEM-EDX elemental mapping of Fe and Al for the fresh Al₂O₃/α-Fe₂O₃=1 catalyst; B) SEM-EDX elemental mapping of Fe and Si for the fresh SiO₂/α-Fe₂O₃=1 catalyst.

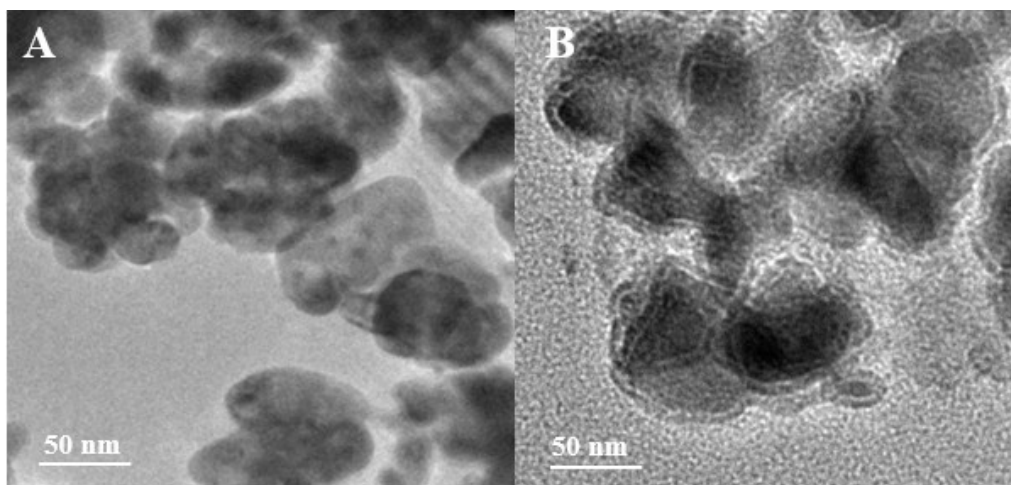


Figure S5 TEM images of the pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst before (A) and after (B) FTS reaction.

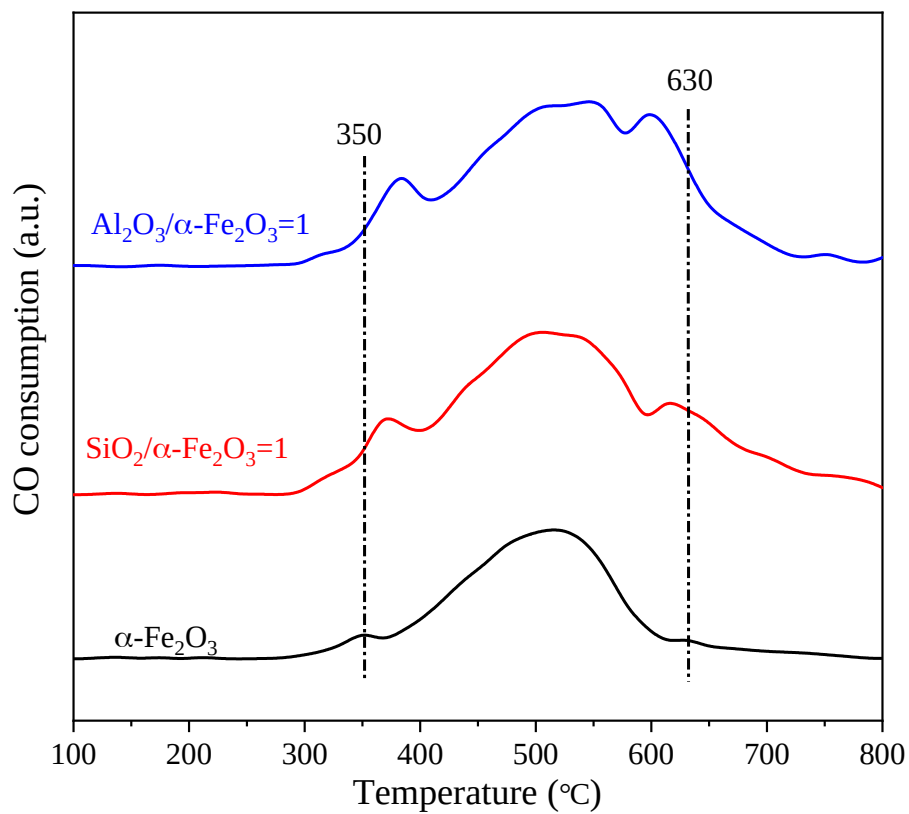


Figure S6 The CO-TPR profiles of various fresh catalysts for CO consumption.

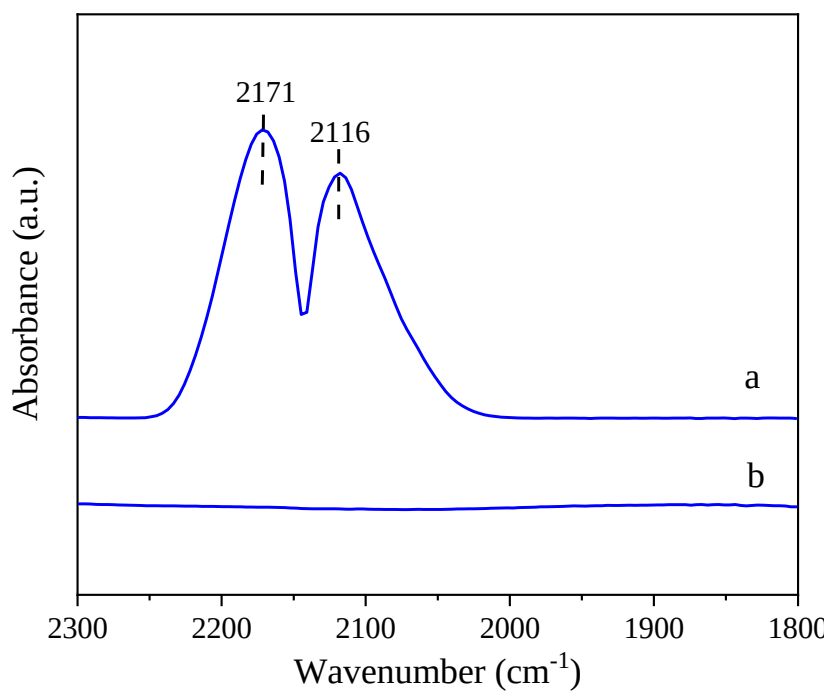
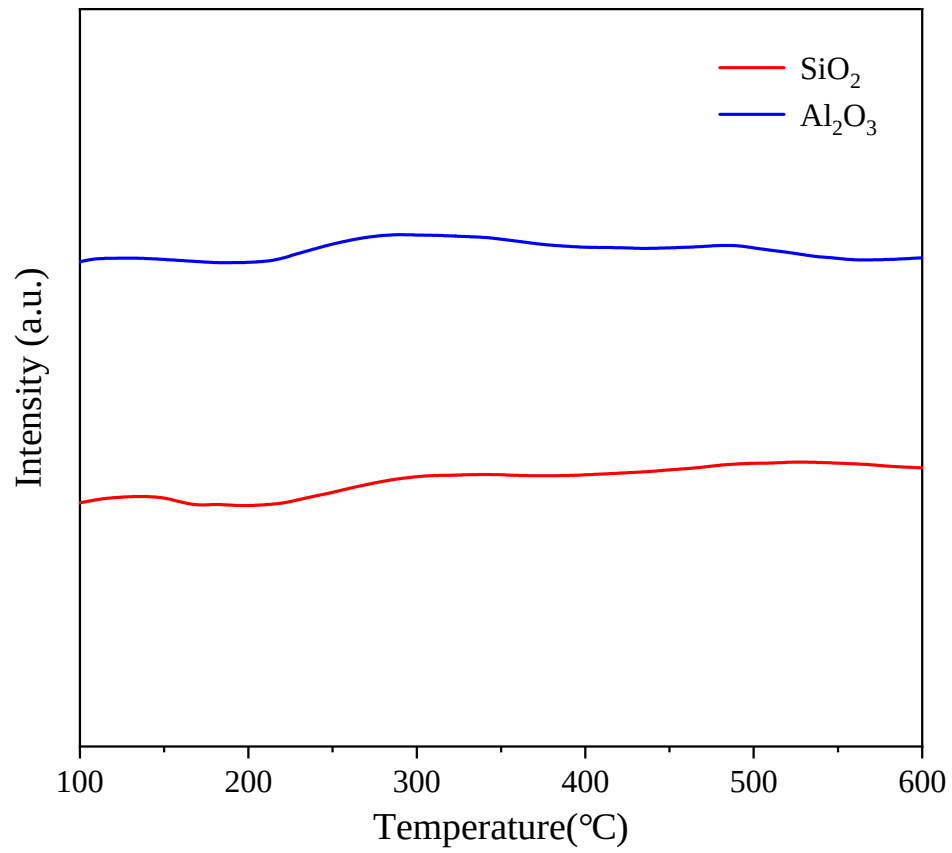


Figure S7 CO-DRIFT spectrum of the Al_2O_3 .

(a)



(b)

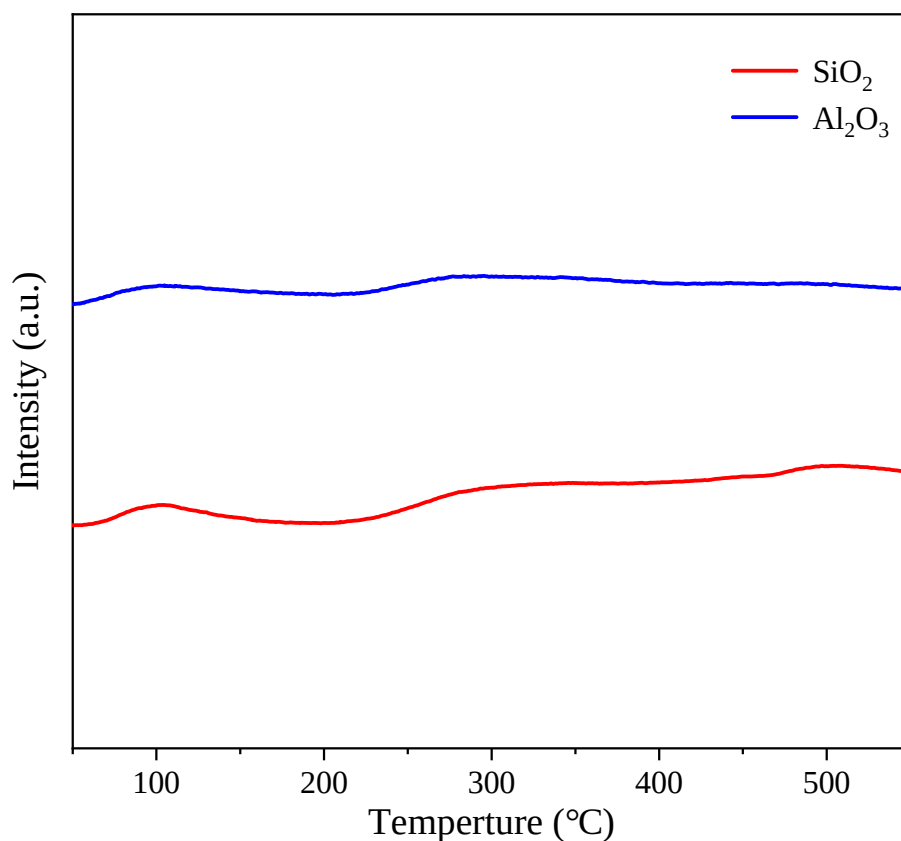


Figure S8 NH_3 -TPD (a) and CO_2 -TPD (b) of SiO_2 and Al_2O_3 .

Table S1. The results of in-situ XPS for Fe 2p 3/2

catalysts	Binding energy (eV)				Iron carbide (atom%)			
	Fe 2p3/2				Fe 2p3/2			
	Fe_2C	Fe_5C_2	Fe_3C	Fe_2O_3	Fe_2C	Fe_5C_2	Fe_3C	Fe_2O_3
$\alpha\text{-Fe}_2\text{O}_3$	706.1	707.0	707.9	709.9	1.2	22.2	15.5	61.1
$\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3=1$	706.7	707.3	708.3	710.4	7.1	19.3	11.7	61.9
$\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3=1$	706.7	707.2	708.2	710.2	13.2	18.0	10.3	58.5

Table S2. The location and content of linear adsorbed CO on different iron carbides from in-situ CO-DRIFT.

catalysts	Wavenumber (cm^{-1})			Content of the adsorbed CO (%)		
	Fe_2C	Fe_5C_2	Fe_3C	Fe_2C	Fe_5C_2	Fe_3C
$\alpha\text{-Fe}_2\text{O}_3$	-	2013	2027	-	63.2	36.8

SiO ₂ /α-Fe ₂ O ₃ =1	2004	2014	2032	18.6	57.3	24.1
Al ₂ O ₃ /α-Fe ₂ O ₃ =1	2006	2016	2033	27.7	49.3	23.0
